

Don't Risk It!: Trajectory-Conditioned Distributional Risk for LiDAR-Only Crowd Navigation

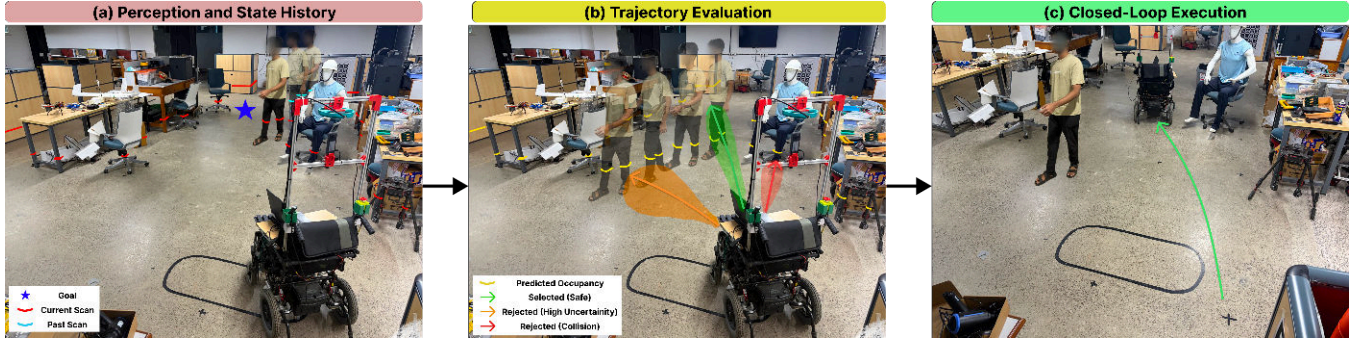


Fig. 1: **Navigation via trajectory-conditioned distributional risk.** (a) **Perception and State History** ($t = 0$ s): Our framework operates strictly from raw 2D LiDAR grid history over the past 1.0 s without object detection or pedestrian tracking, to forecast dynamic scene evolution. (b) **Trajectory Evaluation**: Candidate rollouts are shaded by their predictive variance envelopes ($\pm\sigma$): maneuvers colliding with static obstacles are discarded (**red**); paths attempting aggressive cuts across the pedestrian’s anticipated path, triggering an elevated **risk penalty** (**orange**); and the wide bypass corridor maintains tight predictive uncertainty, certifying safe passage (**green**). (c) **Closed-Loop Execution**: The autonomous wheelchair executes the selected control sequence, avoiding the moving pedestrian.

Abstract—Efficient crowd navigation requires anticipating dynamic collision-free space without relying on fragile object tracking. We propose the Don’t Risk It Planner (DRIP), an end-to-end LiDAR-only planning framework that couples occupancy forecasting with trajectory-conditioned clearance and uncertainty estimation. Operating directly on LiDAR-derived occupancy grid histories, DRIP predicts a distribution over minimum robot–obstacle clearance for candidate control sequences to evaluate collision risk. The model is trained using a risk cost that grounds predictive uncertainty directly in empirical collision outcomes rather than geometric regression alone. In closed-loop experiments with up to 45 pedestrians, DRIP achieves higher success rates with fewer collisions and reduced freezing behavior compared to state-of-the-art baselines. Project page: <https://dont-risk-it-planner.github.io>

I. INTRODUCTION

Safe, efficient crowd navigation requires a mobile robot to anticipate evolving space, as open paths can close and blocked routes may clear. Uncertain pedestrian dynamics and mutual occlusions [1] complicate this reasoning. Crucially, robots must balance anticipatory spatial awareness with calibrated uncertainty estimates to prevent both collisions and over-conservative freezing [2].

Existing approaches separate this problem into pedestrian detection, tracking, motion prediction, and planning. This decomposition introduces a multi-stage perception dependency: detections must be associated over time before pedestrian states can be estimated and their future motion predicted. Besides the additional processing, incomplete observations can make these intermediate estimates unreliable, allowing perception errors to propagate into downstream decisions.

An alternative approach is to forego explicit pedestrian

tracks and directly predict future occupancy maps. For example, [3] presents a self-supervised learning-based approach for spatiotemporal occupancy prediction. More recently, SCOPE [4] extends this direction through stochastic occupancy forecasting and incorporates predicted occupancy and uncertainty into navigation costmaps. These developments provide a basis for anticipatory navigation without an object-centric forecasting pipeline. Nevertheless, bypassing detection and tracking does not eliminate uncertainty in the predicted scene. A central challenge remains: how should uncertainty in occupancy forecasts be represented and learned so that it informs the safety assessment of a candidate robot trajectory?

In this paper, we propose DRIP, an end-to-end, LiDAR-only planning framework that connects occupancy forecasting to trajectory-conditioned clearance and uncertainty estimation. Our central observation is that prediction errors matter differently depending on the trajectory under consideration. An occupancy error far from the robot’s path may have little effect on its safety, whereas a small error near its closest encounter with an obstacle may change whether the trajectory should be selected. Consequently, improving scene-level prediction accuracy alone need not improve the safety assessment most relevant to planning. The broader principle of aligning prediction with downstream risk has also been explored by [5], which learns risk-biased motion forecasts. We pursue this alignment directly from occupancy-level predictions by learning uncertainty in a trajectory-specific safety quantity.

Specifically, as illustrated in Fig. 1, DRIP forecasts scene evolution from a history of LiDAR-derived occupancy grids,

without requiring explicit object detection or tracking. Given a candidate control sequence, a trajectory-conditioned model estimates a distribution over the minimum robot–obstacle clearance across the planning horizon. This representation expresses uncertainty in the closest encounter along the candidate trajectory, rather than only in individual occupancy predictions. The resulting clearance distribution is then mapped to collision risk, providing a common safety measure for comparing candidate motions.

We build upon [6] and train our forecasting model with a risk-calibrated objective that directly aligns the loss with downstream trajectory risk evaluation. By regularizing predicted uncertainty using supervision from safe and unsafe trajectories, our approach couples uncertainty estimation directly to planning safety rather than passive scene prediction. At inference, these risk estimates guide trajectory selection within a sampling-based local planner, balancing collision avoidance with goal progress.

We benchmark our approach in a closed-loop setting in dense environments with up to 45 pedestrians against TEB [7], DWA [8], DRL-VO [9], and SCOPE [4]. In extreme crowding (45 pedestrians), our framework achieves an 85.0% success rate (a 31.8% reduction in navigation failure over state-of-the-art predictive baselines), while reducing completion time by 14.5% and improving velocity by 14.6% compared to SCOPE. Under severe sensor noise, the performance gap widens drastically, cutting traversal time by over 50% (13.4 s vs. 27.0 s) and maintaining an 86.0% success rate.

In summary, the main contributions of this work are:

- **A LiDAR-only predictive planning framework** that couples detection- and tracking-free occupancy forecasting with a trajectory-conditioned distribution over minimum clearance.
- **A risk-calibrated uncertainty-learning objective** that uses supervision from safe and unsafe trajectories to shape predictive variance through its downstream effect on collision-risk estimation.
- **Closed-loop validation in dense crowds and under sensor noise**, demonstrating fewer collisions and less freezing behavior, together with higher navigation success rates than state-of-the-art baselines.

II. RELATED WORK

A. Crowd Navigation in Dynamic Environments

Crowd navigation methods range from geometric collision avoidance, such as velocity obstacles [10], to learned pedestrian interaction models using structural RNNs, graphs, and attention mechanisms [11], [12]. DRL-VO [9] bridges these domains by combining velocity-obstacle rewards with LiDAR observations and pedestrian kinematics. Other methods incorporate risk by constructing local probabilistic risk maps [13] or integrating barrier functions into learned policies [14]. Recent works have also leveraged generative priors, such as VQ-VAEs [15] and Conditional Flow Matching [16], to model multimodal expert trajectories for sampling-based local planning. Within this broader effort, DRIP predicts

candidate motion safety directly from LiDAR histories. By learning clearance uncertainty for each candidate, it evaluates future encounters without requiring the explicit pedestrian-state estimates used by agent-based models and DRL-VO.

B. Occupancy Forecasting for Predictive Navigation

Occupancy maps provide a common representation for anticipating unseen geometry [17], learning social costs [18], and predicting how occupied space evolves. For dynamic navigation, LiDAR-based forecasting offers a way to anticipate scene changes without detecting and tracking individual pedestrians [3]. SOGMP and SOGMP++ [19] capture uncertainty by modeling stochastic future occupancy. SCOPE [4] further develops this approach by incorporating predicted occupancy and uncertainty into navigation costmaps. DRIP builds on this tracking-free approach by connecting scene forecast uncertainty to the safety of each candidate trajectory during training itself. In particular, it learns a distribution over the closest robot–obstacle encounter across the planning horizon. The next-step occupancy forecast supports immediate collision checking, while the full forecast sequence informs this trajectory-conditioned clearance model.

C. Uncertainty-Aware and Distributional Planning

Explicitly accounting for uncertainty allows robots to adjust speeds based on collision confidence [20] and balance safety with goal progress under model uncertainty [21]. For crowd navigation, Ryu and Mehr [22] manage distributional risk via CVaR approximations over pedestrian forecasts. Maximum Mean Discrepancy (MMD) provides another risk assessment tool, recently used to distill stochastic dynamics rollouts [23]. While this approach derives risk from stochastic dynamics, DRIP computes squared MMD over a learned clearance distribution, conditioned on LiDAR histories and candidate controls, to measure how far safety-margin violations depart from an ideal zero-violation case.

D. Learning Predictions for Downstream Risk

Accurate predictions do not always lead to better planning decisions: an error near the robot’s path may matter more than a larger error elsewhere. RAP [5] addresses this mismatch by learning risk-biased forecasts of other agents’ trajectories, so that expected costs approximate a desired risk measure. MonoMPC [6] provides a more direct foundation for our work by learning a control-conditioned minimum-clearance distribution for monocular navigation and supervising its uncertainty through an MMD-based risk model. Building on MonoMPC, DRIP extends clearance-based risk learning to dynamic crowds using temporal LiDAR observations and explicit future occupancy forecasting. Safe/unsafe trajectory labels supervise clearance uncertainty, while trajectory-ranking supervision encourages the occupancy decoder to preserve the relative safety of candidate paths. Together, these objectives connect both predictions to the safety decisions they support.

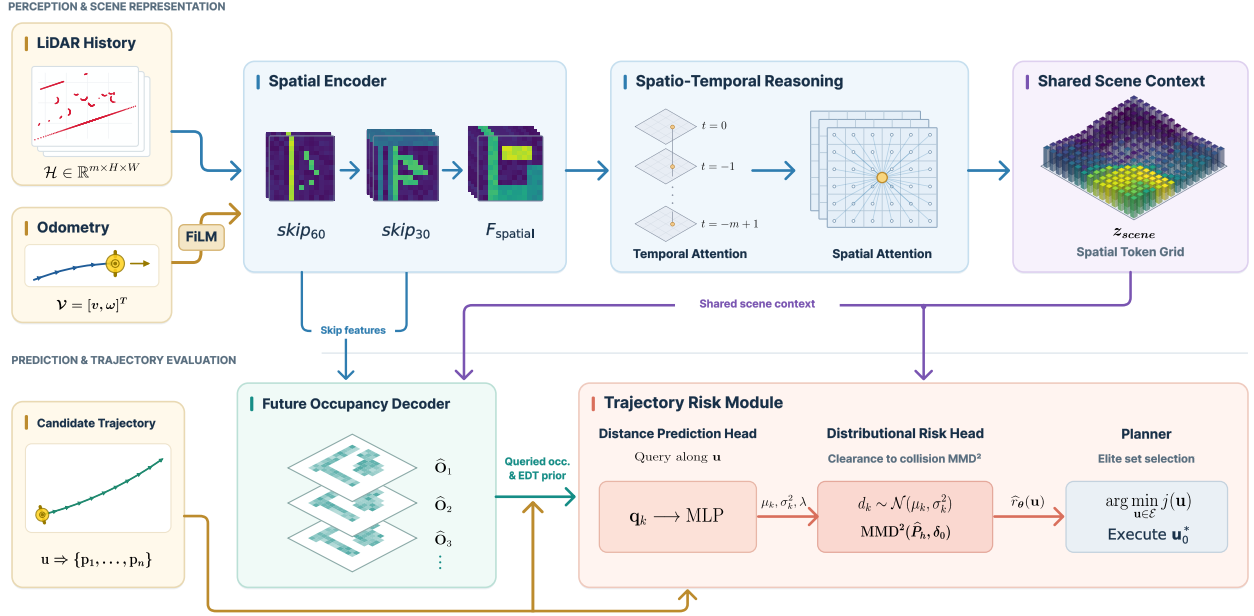


Fig. 2: Overview of DRIP’s LiDAR-only prediction and trajectory-conditioned risk architecture. A history $\mathcal{H}$ of pose-warped BEV occupancy maps is encoded independently by a shared-weight spatial CNN, conditioned on ego velocity $\mathcal{V}$ through FiLM. Spatiotemporal Transformer blocks then reason over the temporal history and spatial BEV tokens to produce a shared scene representation z_{scene} . Two branches use this representation: the occupancy decoder predicts the future evolution of the scene ($\hat{\mathbf{O}}_k$) over a 20-step horizon, while the trajectory-conditioned Distance Head combines scene features with candidate trajectory queries $\mathbf{q}_k$ to predict a stepwise clearance distribution (μ_k, σ_k^2) and a trajectory-specific kernel width λ . These predictions are subsequently used to compute the distributional MMD² risk $\hat{r}_\theta(\mathbf{u})$ for each candidate trajectory $\mathbf{u}$.

III. PROBLEM FORMULATION

We consider a mobile robot navigating toward a goal among static obstacles and moving pedestrians. At each planning cycle, a control sequence $\mathbf{u} = [\mathbf{u}_0^\top, \dots, \mathbf{u}_{n-1}^\top]^\top$ generates an n -step trajectory from the current state $\mathbf{x}_0$ through

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, n-1 \quad (1)$$

where $\mathbf{x}_k \in \mathbb{R}^{n_x}$ and $\mathbf{u}_k \in \mathbb{R}^{n_u}$ are the state and input. Let $\mathbf{p}_k = \mathbf{p}(\mathbf{x}_k(\mathbf{u})) \in \mathbb{R}^2$ denote the resulting planar positions and $\mathbf{g} \in \mathbb{R}^2$ the goal.

The prediction model receives LiDAR occupancy-grid history $\mathcal{H} = (\mathbf{O}_{-m+1}, \dots, \mathbf{O}_0)$ and aligned robot velocity history $\mathcal{V} = (\mathbf{v}_{-m+1}, \dots, \mathbf{v}_0)$, where $\mathbf{v}_j = [v_j, \omega_j]^\top$ contains measured linear and angular velocities. It forecasts occupancy maps $\hat{\mathbf{O}}_{1:n}$ in the current robot body frame. We exploit the higher accuracy of the next-step forecast through the deterministic collision cost

$$j_{\text{occ}}(\mathbf{u}) = \ell_{\text{occ}}(\mathbf{x}_1(\mathbf{u}), \hat{\mathbf{O}}_1) \quad (2)$$

which penalizes footprint overlap with occupied space, holding $\hat{\mathbf{O}}_1$ fixed during optimization.

To assess safety over the full horizon, let $\mathcal{O}_k \subset \mathbb{R}^2$ denote the future obstacle-occupied region. For a circular robot of radius r_{rob} , define

$$d_k(\mathbf{u}, \mathcal{O}_k) = \inf_{\mathbf{q} \in \mathcal{O}_k} \|\mathbf{p}_k - \mathbf{q}\|_2 - r_{\text{rob}}, \quad (3)$$

$$d_{\min}(\mathbf{u}) = \min_{k=1, \dots, n} d_k(\mathbf{u}, \mathcal{O}_k) \quad (4)$$

Positive clearance indicates separation; nonpositive clearance indicates contact or overlap. Uncertain future occupancy

makes $d_{\min}$ random. We learn its conditional law $\mathcal{P}_\theta(\cdot \mid \mathcal{H}, \mathcal{V}, \mathbf{u})$ and evaluate the risk

$$\hat{r}_\theta(\mathbf{u}) = \rho(\mathcal{P}_\theta(\cdot \mid \mathcal{H}, \mathcal{V}, \mathbf{u})) \quad (5)$$

where θ denotes learned parameters and ρ is a distribution-based risk functional. We use the MMD construction in Section IV to model this.

For a candidate control sequence $\mathbf{u}$, the goal-reaching and smoothness costs are

$$j_{\text{goal}} = \|\mathbf{p}_n - \mathbf{g}\|_2^2, \quad j_{\text{smooth}} = \sum_{k=0}^{n-1} \|\mathbf{u}_k - \mathbf{u}_{k-1}\|_2^2 \quad (6)$$

where $\mathbf{u}_{-1}$ is the previously applied input. We solve

$$\begin{aligned} \mathbf{u}^* &= \arg \min_{\mathbf{u}} w_g j_{\text{goal}}(\mathbf{u}) + w_s j_{\text{smooth}}(\mathbf{u}) \\ &\quad + w_o j_{\text{occ}}(\mathbf{u}) + w_r \hat{r}_\theta(\mathbf{u}) \\ \text{s.t. } &\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k), \quad \mathbf{x}_0 = \mathbf{x}_{\text{current}}, \\ &\mathbf{u}_k \in \mathcal{U}, \quad \mathbf{x}_{k+1} \in \mathcal{X}. \end{aligned} \quad (7)$$

The constraints hold for $k = 0, \dots, n-1$. The non-negative weights balance the four costs, and $\mathcal{U}, \mathcal{X}$ encode admissible inputs and states. A sampling-based planner approximately solves this problem, executes $\mathbf{u}_0^*$, and replans with updated observations.

IV. METHODS

Fig. 2 summarizes our prediction pipeline. A shared spatial encoder processes the LiDAR history $\mathcal{H}$, with the aligned robot velocity history $\mathcal{V}$ conditioning its feature channels through FiLM (feature-wise linear modulation) layers. Temporal attention combines features across observations, and

spatial attention captures relationships across the bird’s-eye-view (BEV) grid, yielding the shared scene representation $\mathbf{z}_{scene}$.

Two connected branches use this representation. The occupancy decoder combines $\mathbf{z}_{scene}$ with future-time embeddings and multi-scale skip features to forecast occupancy maps $\hat{\mathbf{O}}_{1:n}$. The trajectory-conditioned branch queries scene features, occupancy forecasts, and their Euclidean distance-transform priors along a candidate trajectory. These queries combine scene context with local free-space geometry to predict stepwise clearance means and variances, together with a trajectory-specific kernel bandwidth. For each sampled clearance rollout, we take the minimum over the horizon to obtain a sample of trajectory-level minimum clearance. The risk module transforms these samples into safety-margin violations and compares their empirical distribution with a point mass at zero. It uses squared MMD with the predicted bandwidth λ to produce the risk estimate $\hat{r}_\theta(\mathbf{u})$. During training, we use supervision from ground-truth occupancy to supervise $\hat{\mathbf{O}}_{1:n}$. Moreover, following [6], we compare the predicted risk $\hat{r}_\theta(\mathbf{u})$ with the ground-truth estimate and use the error to align the uncertainty prediction.

We next present the building blocks of our pipeline, followed by the training procedure and online planner.

A. Spatiotemporal Scene Encoding and Occupancy Forecasting

1) *Shared scene representation*: The observation history $\mathcal{H}$ contains m LiDAR-derived BEV occupancy grids. Past scans are warped into the current robot body frame using measured ego-motion. A spatial encoder with shared weights processes each grid through convolutional stages at 60×60 , 30×30 , and 15×15 resolutions. At the deepest stage, FiLM conditions the feature channels on the linear and angular velocity history $\mathcal{V}$. This history provides the robot-motion context aligned with the input LiDAR frames.

Temporal self-attention combines the m observations independently at each spatial cell. Spatial self-attention then processes the latest-time feature grid to relate information across the scene. The resulting scene representation $\mathbf{z}_{scene}$ encodes observed geometry and motion cues without requiring pedestrian detections or persistent object identities.

2) *Occupancy forecasts for collision checking and clearance prediction*: A convolutional decoder combines $\mathbf{z}_{scene}$, learned future-time embeddings $\mathbf{e}_k$, and multi-scale features $\mathbf{F}_0$ from the latest observation to predict

$$\hat{\mathbf{O}}_k = f_{\text{dec}}(\mathbf{z}_{scene}, \mathbf{F}_0, \mathbf{e}_k), \quad k = 1, \dots, n \quad (8)$$

where each $\hat{\mathbf{O}}_k \in [0, 1]^{h_{\text{bev}} \times w_{\text{bev}}}$ is a soft-occupancy grid. Skip connections from $\mathbf{F}_0$ retain fine spatial detail during up-sampling. We also construct a Euclidean distance-transform grid EDT_k from each forecast to provide a local free-space distance prior.

The first forecast $\hat{\mathbf{O}}_1$ supplies the direct next-step collision check in (2). For each candidate, the check evaluates the robot footprint at its first predicted state using this fixed map. The complete forecast sequence and its distance-transform

priors also provide geometric context for the clearance model. Querying them along all future waypoints supports the minimum-clearance distribution and full-horizon risk defined in Section III.

B. From Candidate Controls to Clearance and Risk

1) *Trajectory-conditioned clearance marginals*: For a candidate $\mathbf{u}$, the dynamics in (1) determine the waypoint positions $\mathbf{p}_k = \mathbf{p}(\mathbf{x}_k(\mathbf{u}))$, $k = 1, \dots, n$. For querying, these positions are expressed in the same frame as the occupancy grids. At each waypoint, we construct the feature vector,

$$\mathbf{q}_k = [\mathbf{z}_k^\top, \hat{\mathbf{O}}_k(\mathbf{p}_k), \text{EDT}_k(\mathbf{p}_k), \mathbf{f}_{\text{skip}}(\mathbf{p}_k)^\top]^\top \quad (9)$$

where $\mathbf{z}_k \in \mathbb{R}^d$ is the spatiotemporal latent token extracted at waypoint $\mathbf{p}_k$, $\hat{\mathbf{O}}_k(\mathbf{p}_k)$ denotes the forecasted occupancy probability obtained via bilinear interpolation of the predicted k -th step BEV occupancy map at position $\mathbf{p}_k$, $\text{EDT}_k(\mathbf{p}_k)$ is the Euclidean distance transform querying distance to static boundaries, and $\mathbf{f}_{\text{skip}}(\mathbf{p}_k)$ represents raw spatial coordinate skip-features.

A shared multilayer perceptron maps $\mathbf{q}_k$ to a mean μ_k and a positive variance σ_k^2 , giving the conditional marginal model

$$d_k \mid \mathcal{H}, \mathcal{V}, \mathbf{u} \sim \mathcal{N}(\mu_k, \sigma_k^2), \quad k = 1, \dots, n \quad (10)$$

These are distributions of the footprint-adjusted clearance in (3); the robot radius is therefore already accounted for. We suppress the conditioning arguments of μ_k and σ_k for readability. A second head pools the query features over the horizon to predict a positive kernel bandwidth $\lambda_\theta(\mathbf{u})$ for the risk computation below. Its dependence on $\mathcal{H}$ and $\mathcal{V}$ is likewise implicit.

2) *Minimum-clearance distribution*: We obtain samples of trajectory-level minimum clearance by first sampling the stepwise marginals and then taking the minimum within each sampled rollout:

$$d_k^{(s)} = \mu_k + \sigma_k \epsilon_k^{(s)}, \quad \epsilon_k^{(s)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1) \quad (11a)$$

$$d_{\min}^{(s)}(\mathbf{u}) = \min_{k=1, \dots, n} d_k^{(s)}, \quad s = 1, \dots, n_s \quad (11b)$$

3) *Distributional risk from safety-margin violations*: Let $d_{\text{safe}} > 0$ denote the required clearance buffer beyond the robot footprint. For a candidate $\mathbf{u}$, the random variable $h(\mathbf{u}) = \max(0, d_{\text{safe}} - d_{\min}(\mathbf{u}))$ measures how much this buffer is violated. The uncertainty in minimum clearance therefore induces a distribution over safety-buffer violations, which we denote by $P_h(\mathbf{u})$.

To evaluate this distribution, we transform the n_s minimum-clearance samples from (11b) and form their empirical law:

$$h^{(s)}(\mathbf{u}) = \max(0, d_{\text{safe}} - d_{\min}^{(s)}(\mathbf{u})) \quad (12)$$

$$\hat{P}_h(\mathbf{u}) = \frac{1}{n_s} \sum_{s=1}^{n_s} \delta_{h^{(s)}(\mathbf{u})} \quad (13)$$

Each sample is zero when its rollout respects the buffer and positive by the amount of the buffer shortfall otherwise. Together, these samples form an empirical approximation

of $P_h(\mathbf{u})$. Here, δ_a denotes a unit point mass at a , so $\hat{P}_h$ assigns equal weight $1/n_s$ to each sampled violation. The hat identifies this sample-based representation. An ideal safe trajectory has zero violation for every realization, giving the target distribution δ_0 .

We measure discrepancy from this target using squared MMD with a Laplacian kernel, $k_\lambda(a, b) = \exp(-|a - b|/\lambda)$. For fixed observations and candidate $\mathbf{u}$, write $\lambda = \lambda_\theta(\mathbf{u})$. Since the target is a point mass, the empirical risk has the explicit form

$$\hat{r}_\theta(\mathbf{u}) = \text{MMD}_{k_\lambda}^2(\hat{P}_h(\mathbf{u}), \delta_0) \quad (14)$$

$$= \frac{1}{n_s^2} \sum_{s,t=1}^{n_s} \exp(-|h^{(s)} - h^{(t)}|/\lambda) - \frac{2}{n_s} \sum_{s=1}^{n_s} \exp(-h^{(s)}/\lambda) + 1 \quad (15)$$

Equation (15) evaluates the empirical MMD directly from the violation samples. In the notation of (5), ρ includes the transformation from clearance to violation and uses the candidate's predicted bandwidth.

C. Risk-Calibrated Training

We build upon [6] to develop a training pipeline that regularizes uncertainty not only through the forecasting error but also through how it affects the downstream planning risk. This is done in two ways: (i) by first aligning predicted risk during training with the ground-truth risk and (ii) by a ranking loss that differentiates between safe and unsafe trajectory proposals.

1) *Stage 1: occupancy forecasting*: We first optimize the spatial encoder, attention modules, and occupancy decoder using ground-truth future occupancy grids $\mathbf{O}_{1:n}^{\text{gt}}$:

$$\mathcal{L}_{\text{recon}} = (\mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{Dice}})(\hat{\mathbf{O}}_{1:n}, \mathbf{O}_{1:n}^{\text{gt}}) \quad (16)$$

Focal and Dice loss functions address the imbalance between occupied and free cells. To emphasize changes in occupancy, reconstruction supervision uses spatial weights

$$\mathbf{W}_k(\mathbf{p}) = 1 + \beta_{\text{dyn}} \mathbb{I}[\mathbf{O}_k^{\text{gt}}(\mathbf{p}) \neq \mathbf{G}_0(\mathbf{p})] \quad (17)$$

with $\beta_{\text{dyn}} = 4$. The current and future grids are expressed in the same frame. This weighting emphasizes cells that become occupied or free, reducing the dominance of unchanged background geometry in the training signal.

2) *Stage 2: clearance and risk supervision*: We freeze the spatial encoder and attention modules, retain reconstruction supervision for the occupancy decoder, and optimize

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{recon}} + \beta_{\text{dist}} \mathcal{L}_{\text{dist}} + \beta_{\text{risk}} \mathcal{L}_{\text{risk}} + \beta_{\text{haz}} \mathcal{L}_{\text{haz}} \quad (18)$$

where the nonnegative β coefficients weight the objectives. *Clearance likelihood*: For each sampled candidate, ground-truth stepwise clearances $d_k^{\text{gt}}(\mathbf{u})$ are computed using the same robot footprint convention as (3). Omitting an additive constant, the Gaussian negative log-likelihood is

$$\mathcal{L}_{\text{dist}} = \frac{1}{2n} \sum_{k=1}^n [(d_k^{\text{gt}} - \mu_k)^2 / \sigma_k^2 + \log \sigma_k^2] \quad (19)$$

The residual term penalizes inaccurate means relative to the predicted variance, while the logarithmic term discourages explaining every error by increasing the variance.

Risk-based trajectory supervision: The ground-truth minimum clearance defines the binary label

$$y(\mathbf{u}) = \mathbb{I}[\min_{1 \leq k \leq n} d_k^{\text{gt}}(\mathbf{u}) < d_{\text{safe}}] \quad (20)$$

where $y = 1$ denotes a safety-buffer violation. A lightweight classifier g_ϕ takes the empirical MMD score and violation statistics as input:

$$\ell(\mathbf{u}) = g_\phi(\hat{r}_\theta(\mathbf{u}), \bar{h}, \sigma_h, h_{\text{max}}), \quad (21a)$$

$$\mathcal{L}_{\text{risk}} = \text{BCEWithLogits}(\ell(\mathbf{u}), y(\mathbf{u})) \quad (21b)$$

Here, $\bar{h}$ is the average constraint violation residual, σ_h is the standard deviation of these residuals, and h_{max} is the worst-case constraint violation across the sampled rollouts. These statistics summarize the average severity, variability, and largest violation associated with the candidate trajectory. The classifier parameters ϕ are used only during training. Gradients pass through the violation statistics and MMD computation to train the clearance head and bandwidth predictor. Consequently, the uncertainty receives supervision through its effect on distinguishing safe and unsafe trajectories, in addition to the likelihood objective.

a) *Auxiliary Hazard Supervision*: To condition the future occupancy decoder beyond passive pixel reconstruction, we introduce an auxiliary hazard-margin loss evaluated over candidate path queries. For each trajectory $\mathbf{u}$, we bilinearly interpolate the soft occupancy probability at each waypoint and compute the scalar hazard cost $c(\mathbf{u})$ by averaging the highest values over a 0.5 s window. For trajectory pairs where path i is strictly safer than path j in ground truth ($c_{\text{gt}}(\mathbf{u}^{(i)}) < c_{\text{gt}}(\mathbf{u}^{(j)})$), we penalize the decoder if the predicted hazard separation falls below margin $m_{\text{haz}} = 0.05$:

$$\mathcal{L}_{\text{haz}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max(0, c_{\text{pred}}(\mathbf{u}^{(i)}) - c_{\text{pred}}(\mathbf{u}^{(j)}) + m_{\text{haz}}). \quad (22)$$

This directly conditions the occupancy decoder, focusing gradient updates on navigable path boundaries rather than passive background cells.

Gradient routing: During Stage 2, scene features, skip features, occupancy values, and distance-transform priors are detached before they enter the trajectory querying branch. The likelihood loss $\mathcal{L}_{\text{dist}}$ trains the trajectory-query and clearance modules. The risk loss $\mathcal{L}_{\text{risk}}$ also updates these modules and additionally trains the bandwidth predictor and classifier g_ϕ . Neither loss updates perception through this branch. The occupancy decoder is updated by $\mathcal{L}_{\text{recon}}$ and $\mathcal{L}_{\text{haz}}$ through its undetached outputs. The encoder and attention modules remain frozen. This routing preserves the shared scene representation while allowing both downstream objectives to refine their respective branches.

D. Sampling-Based Optimization and Learned Model Inference

We solve (7) using DWA-style trajectory sampling. At each cycle, the histories $(\mathcal{H}, \mathcal{V})$ produce the shared representation

$\mathbf{z}_{scene}$, occupancy forecasts $\hat{\mathbf{O}}_{1:n}$, and distance-transform priors once. We sample n_c admissible control sequences and roll them out from $\mathbf{x}_0$ subject to the dynamics and state limits. Using the cached perception outputs, we evaluate next-step occupancy costs and trajectory-conditioned risk in a batch, with shared clearance-noise samples and fixed learned parameters. We adopt a two-stage approach. We first select a subset of sampled trajectories with the lowest safety cost (one-step occupancy+risk, recall (6)). We then evaluate the full cost over these subset to choose the lowest cost trajectory for execution.

V. VALIDATION AND BENCHMARKING

We evaluate DRIP against state-of-the-art classical, optimization-based, and learning-based navigation planners in simulated indoor environments with dynamic crowds. Our experiments investigate two primary questions:

- 1) **Scalability Across Crowd Densities:** Does trajectory-conditioned distributional risk improve navigation success and improve robot behavior in high pedestrian density cases?
- 2) **Robustness to Sensor Degradation:** Does evaluating continuous clearance distributions maintain path consistency under LiDAR range noise where classical costmap planners degrade?

A. Implementation and Experimental Setup

a) *Robot Platform and Simulation Environment:* We use a differential-drive Turtlebot2 mobile base with a forward-facing 2D LiDAR modeled after the Hokuyo UTM-30LX, with a 270° field of view. We conduct simulated evaluations in the PEDSIM Simulator [24], evaluating the Lobby world with 35 and 45 active pedestrians.

b) *Data Collection and Trajectory Supervision:* To train the model, we construct a hybrid dataset comprising:

- **Synthetic Simulation ($\sim 500k$):** Room, corridor, and intersection topologies simulated via PySocialForce [25].
- **Real-World Data ($\sim 750k$):** Extracted from ≈ 75 hours of HuRoN [26] and ≈ 5 hours of SCAND [27] logs to capture authentic human motion and sensor noise.

To supervise our trajectory-conditioned risk and ranking objectives, raw LiDAR point clouds from these datasets are ego-motion compensated and projected into consecutive BEV grids registered to the latest robot pose.

c) *Baselines and Evaluation Protocol:* We benchmark DRIP against four leading navigation policies under identical dynamic limits. Crucially, while DRIP, DWA, TEB, and SCOPE operate strictly from raw 2D LiDAR observations without explicit agent tracking, DRL-VO assumes privileged dynamic state inputs. Performance is measured using four standard criteria: (i) **Success Rate (%)** (reaching the goal without collision or timeout, $T_{\max} = 90$ s); (ii) **Average Time (s)** to completion; (iii) **Average Path Length (m)**; and (iv) **Average Speed (m/s)**.

B. Validation of the Learned Risk Model

To validate our risk-calibrated training objective, we measure the best-of- K residual between sampled trajectory clearances and the ground-truth minimum clearance. As illustrated in Fig. 3, our risk-calibrated model produces a significantly lower median error and tighter inter-quartile range than the NLL-only baseline. Grounding uncertainty estimation in downstream collision risk directly improves the accuracy of the learned distances, ensuring the predicted clearance boundaries closely match actual pedestrian encounters, ensuring reliable navigation even in dense pedestrian traffic.

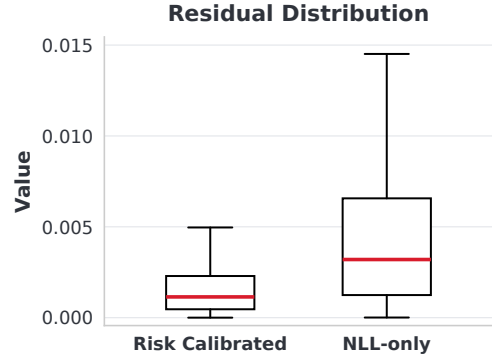


Fig. 3: Box plot of best-of- K clearance residuals relative to privileged ground truth across dynamic evaluation episodes. The risk-calibrated model ((18)) yields a lower median error and narrower spread than the NLL-only baseline ((18) without $\mathcal{L}_{risk}$, $\mathcal{L}_{haz}$), demonstrating well-calibrated predictive variance that tightly bounds actual obstacle encounters.

C. Simulation Benchmarks: Density Scalability and Noise Robustness

Table I evaluates closed-loop navigation performance and sensor noise resilience in the simulated Lobby world under high crowd densities (35 and 45 active pedestrians).

a) *Nominal Navigation and Crowd Scalability:* Under dense crowd interaction, classical geometric planners like DWA and TEB suffer a performance degradation. SCOPE partially mitigates collisions via predicted occupancy, but translating stochastic forecasts into static Gaussian inflation layers induces excessive conservatism, leading to the slowest completion times. Although the privileged DRL-VO achieves high forward speeds using ground-truth agent tracking, its lack of explicit metric clearance constraints leads to collisions. In contrast, DRIP achieves the highest overall success rate because the deterministic Euclidean distance transform penalty prevents collisions against static walls while the learned MMD risk navigates dynamic human flow.

b) *Resilience Under Sensor Degradation:* Injecting independent Gaussian noise ($\sigma = 0.05$ m) into raw LiDAR ranges exposes fundamental failure modes in costmap-based navigation. DWA and TEB have a significant drop in performance due to added noise, as it causes DWA to repeatedly trigger emergency braking and destabilizes TEB’s elastic-band optimization. SCOPE filters some point noise, but range uncertainty compounds costmap inflation. Conversely, DRIP maintains robust performance under identical corruption,

Density	Planner	Nominal Navigation (100 runs each)				Sensor Noise Degradation ($\sigma = 0.05$ m) (50 runs each)			
		Succ. (%) $\uparrow$	Time (s) $\downarrow$	Path (m) $\downarrow$	Vel. (m/s) $\uparrow$	Succ. (%) $\uparrow$	Time (s) $\downarrow$	Path (m) $\downarrow$	Vel. (m/s) $\uparrow$
35 ped	DRLVO [†]	74.0	11.5	5.39	0.460	—	—	—	—
	DWA	79.0	13.5	5.08	0.355	38.0	31.8	7.33	0.195
	SCOPE	80.0	14.8	5.42	0.365	66.0	26.3	6.10	0.227
	TEB	71.0	12.3	5.46	0.413	30.0	40.2	18.09	0.368
	Ours	87.0	12.1	4.83	0.394	82.0	12.9	4.67	0.357
45 ped	DRLVO [†]	82.0	11.5	5.42	0.455	—	—	—	—
	DWA	74.0	14.7	5.47	0.352	52.0	28.8	5.60	0.175
	SCOPE	78.0	15.2	5.09	0.328	64.0	27.0	5.53	0.204
	TEB	76.0	12.5	5.38	0.413	20.0	46.5	21.01	0.312
	Ours	85.0	13.0	4.82	0.376	86.0	13.4	4.87	0.356

TABLE I: Closed-loop navigation performance and sensor noise robustness in the simulated Lobby world with 35 and 45 pedestrians. Higher success rate ($\uparrow$) and velocity ($\uparrow$) are better; lower completion time ($\downarrow$) and path length ($\downarrow$) are better. **Best** and **Second-best** results are highlighted. ([†]DRL-VO utilizes privileged pedestrian tracking/velocities and is excluded from direct sensor noise testing).

almost matching its noise-free baseline—while preserving direct trajectories. This noise immunity is due to our trajectory risk, as our distance head predicts higher variance σ_k^2 for these noisy point clouds, which then penalizes the closed-form MMD risk $\hat{r}_\theta(\mathbf{u})$ via (15). Rather than emergency braking, DRIP hedges against observation uncertainty by committing to less crowded paths while maintaining continuous forward progress.

D. Qualitative Forecasting Analysis

To understand the planner’s high success rate, we qualitatively evaluate the underlying spatiotemporal forecasting module. Fig. 4 illustrates the model’s ability to anticipate scene evolution strictly from LiDAR histories. Despite lacking explicit pedestrian tracking or data association, the network accurately forecasts the continuous motion of dynamic obstacles while preserving static boundary structures. This tracking-free prediction directly enables the distributional risk assessment used by the local planner.

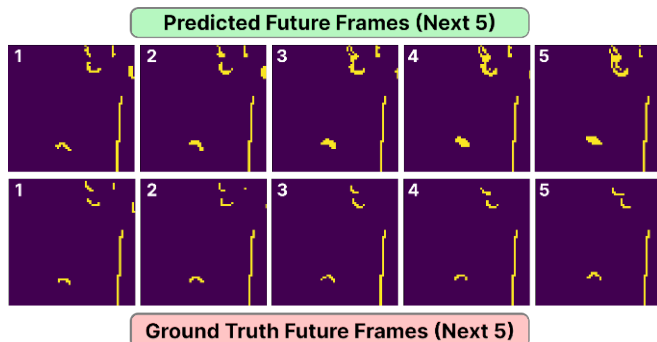


Fig. 4: Visualization of the occupancy forecasting. *Top row*: The first 5 steps of the predicted future occupancy maps ($\hat{\mathbf{O}}_k$), visualized as thresholded binary maps, taken from the full 20-step planning horizon. *Bottom row*: The corresponding ground-truth future occupancy maps ($\mathbf{O}_k^{\text{gt}}$). The network successfully anticipates the continuous motion of dynamic obstacles and maintains static boundary structures, closely matching the ground truth without relying on explicit pedestrian tracking.

E. Runtime Analysis

To verify real-time feasibility for closed-loop deployment, we benchmark the complete inference and planning pipeline on a mobile workstation equipped with an NVIDIA GeForce RTX 3060 Laptop GPU. The local planner targets a strict

10 Hz control loop while evaluating N_{cand} kinematically unrolled trajectory candidates over a (2.0 s) planning horizon. The entire decision loop executes in about 70 ms, which is well within the 100 ms cycle threshold. This computational headroom ensures consistent real-time re-planning while leaving ample processing budget for onboard state estimation and low-level motor actuation.

F. Real-World Hardware Deployment

To demonstrate real-world applicability and sim-to-real transfer, we deploy our navigation framework on two distinct robotic platforms with differing form factors and sensing modalities. For both platforms, the complete perception and planning pipeline was executed entirely onboard using the mobile workstation described in Section V-E:

- 1) **Pioneer 3-DX Mobile Robot**: A compact differential-drive research platform equipped with a 2D RPLIDAR A3 scanner.
- 2) **Autonomous Wheelchair**: A custom full-scale smart wheelchair platform (pictured in Fig. 1) equipped with a Livox Mid-360 3D LiDAR. 3D point clouds are filtered, ground-segmented, and projected into dynamic 2D BEV occupancy grids registered to the latest robot pose.

a) Deployment Scenarios and Results: We conducted field evaluations across two demanding indoor environments: a cluttered laboratory space with narrow passages, and an academic corridor characterized by bi-directional pedestrian flows and unpredictable group motion. Across 20 total closed-loop trials (spanning path lengths of 15 to 35 m) where humans acted as unscripted dynamic obstacles, DRIP achieved an 85.0% success rate (17 out of 20 runs) without requiring human intervention. In open corridors, the planner proactively committed to wider bypass paths around walking pedestrians well before mutual deceleration thresholds. In tight hallway bottlenecks, the decoupled clearance safety floor prevented the base from grazing walls while the learned MMD risk safely moderated forward speed through moving crowd pockets, avoiding freezing deadlock.

VI. CONCLUSION AND FUTURE WORK

In this work, we showed that conventional crowd navigation paradigms exhibit fundamental failure modes in dense

environments: classical geometric costmaps inflate dynamic obstacles into impassable barriers, while end-to-end learning methods either lack formal safety margins or depend on privileged multi-agent tracking pipelines that fail in crowded scenes. We addressed these limitations by introducing DRIP, a framework that formulates safe crowd navigation as a trajectory-conditioned distributional risk evaluation problem directly over raw LiDAR observations.

Closed-loop benchmarks in the simulation demonstrate that DRIP maintains high success rates in crowded scenes, outperforming both classical planners (DWA, TEB) and privileged tracking-based reinforcement learning (DRL-VO). Furthermore, evaluation under synthetic sensor degradation reveals that when exposed to LiDAR coordinate jitter, DRIP preserves safe paths without much performance drop, whereas other methods are much more brittle. Finally, real-world deployments on multiple robots confirm safe planning across various real-world scenes.

A primary limitation of our current framework is its reliance on a discrete set of unrolled trajectory primitives, which can restrict maneuver expressiveness in complex, topologically confined geometries. Future work will investigate integrating continuous generative trajectory priors directly with our distributional MMD risk metric to combine expressive path generation with rigorous safety certification.

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